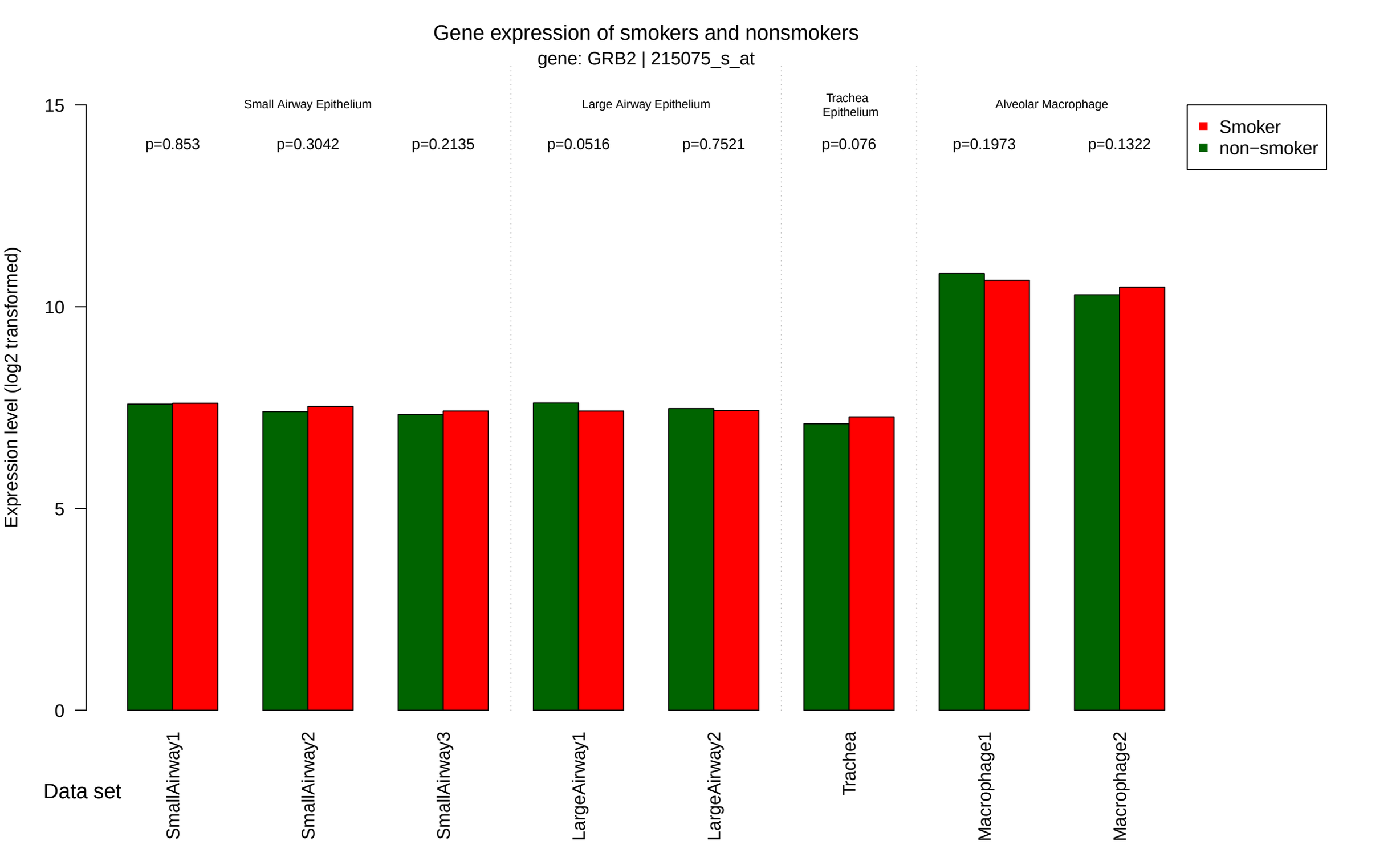
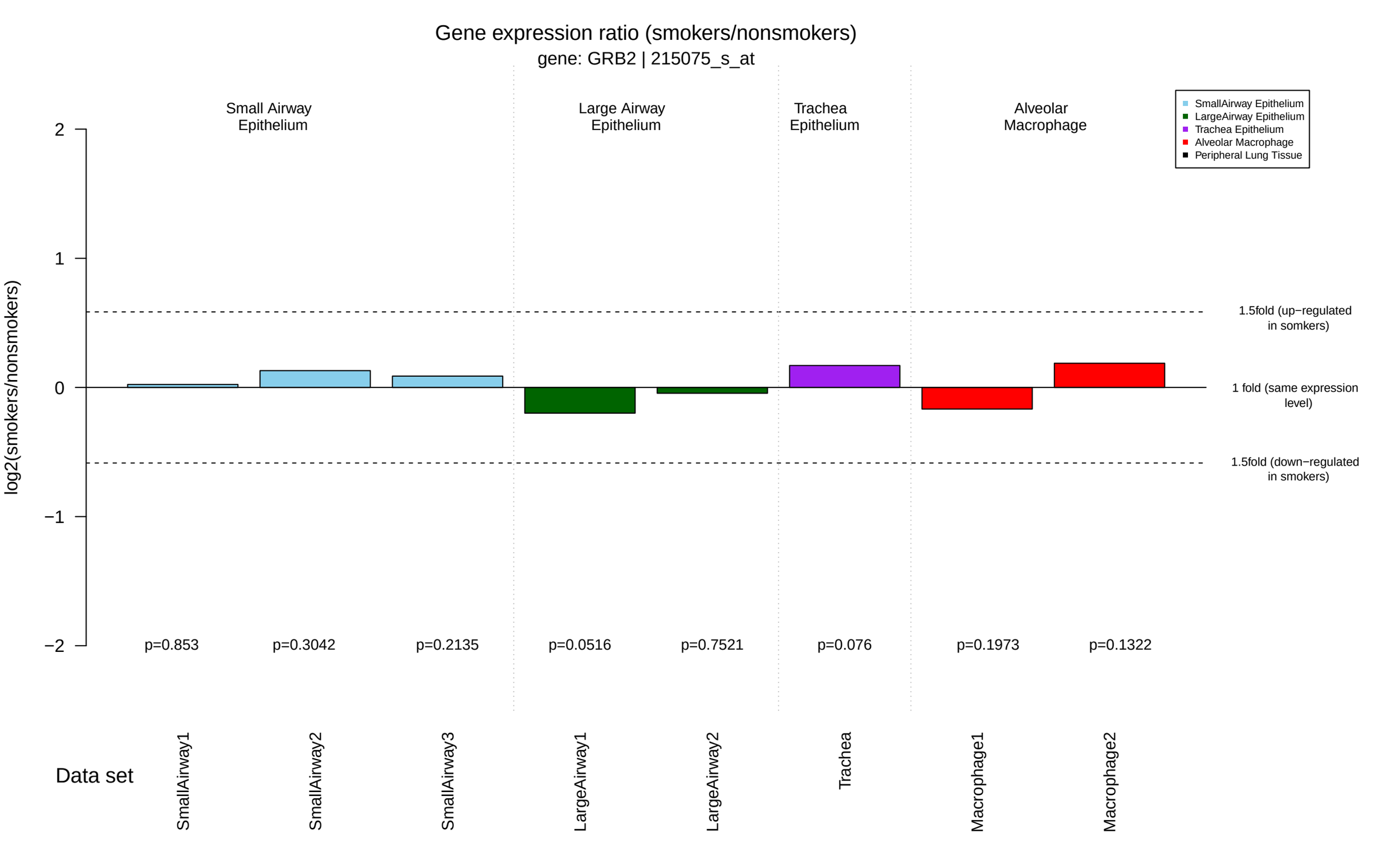
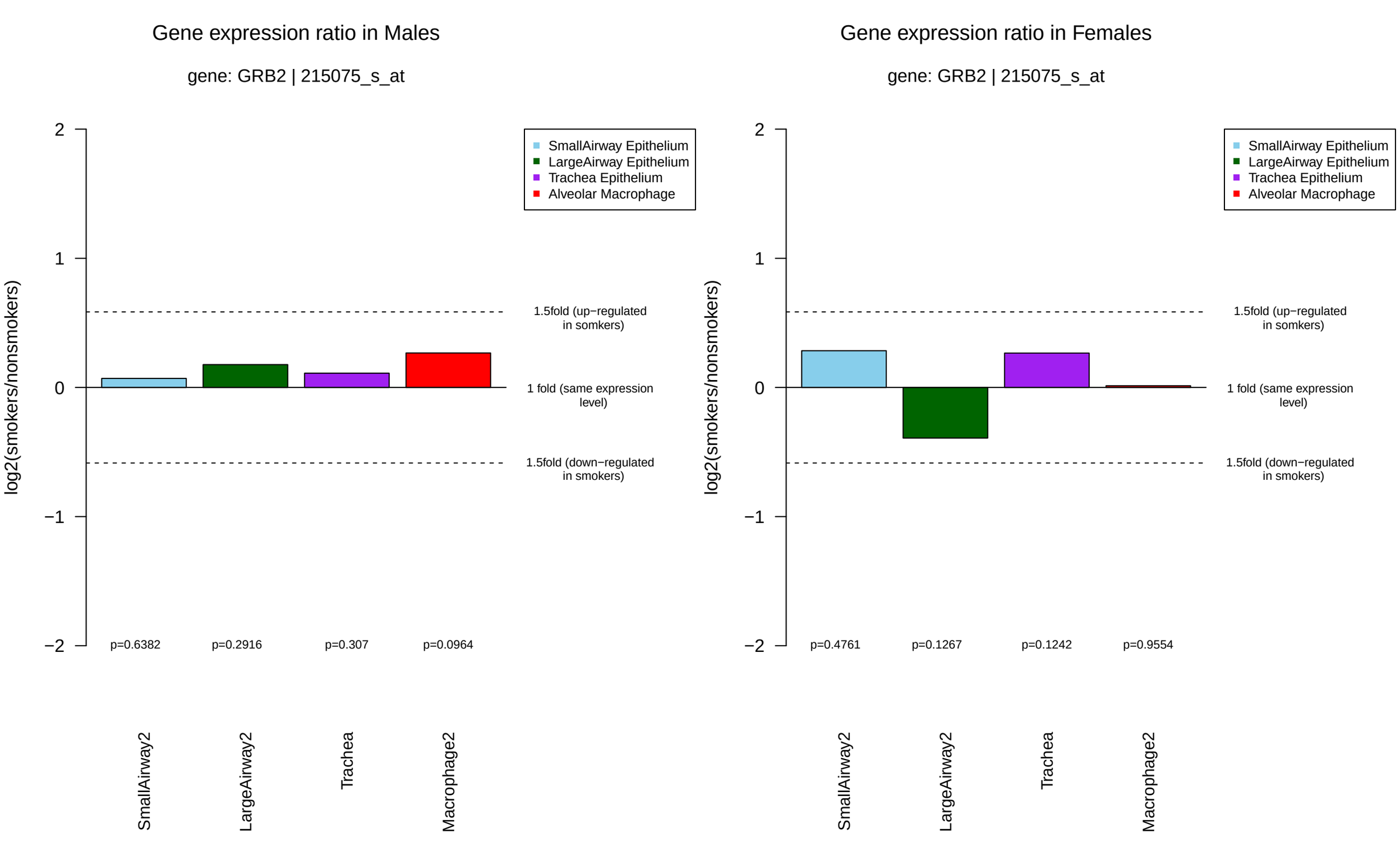
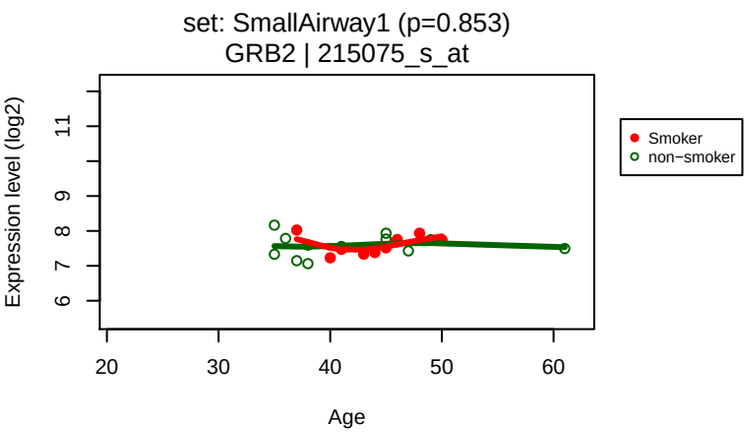
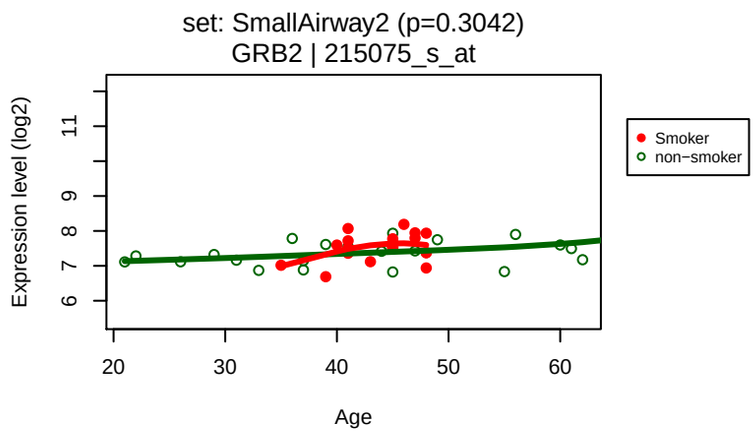
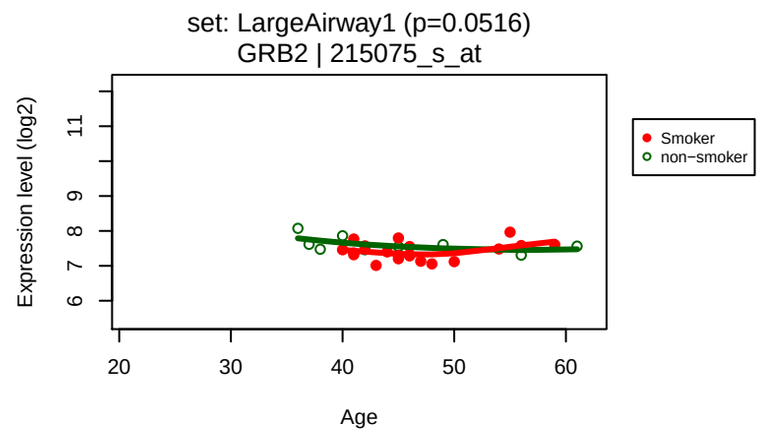

Gene expression of smokers and nonsmokers
gene: GRB2 | 215075_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.853
p=0.3042
p=0.2135
p=0.0516
p=0.7521
p=0.076
p=0.1973
p=0.1322
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

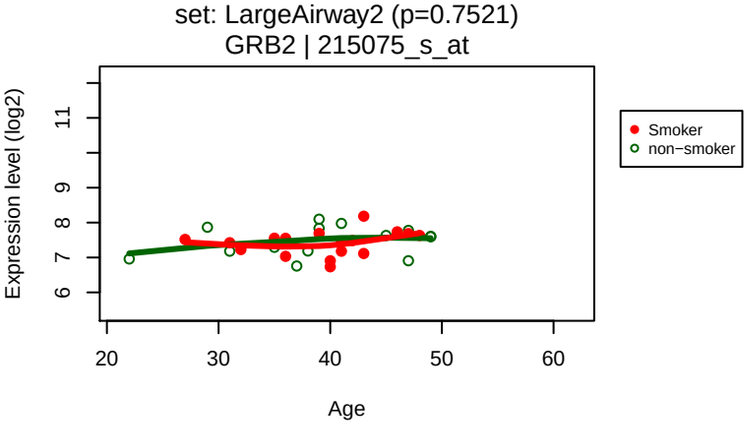

Gene expression ratio (smokers/nonsmokers)
gene: GRB2 | 215075_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.853
p=0.3042
p=0.2135
p=0.0516
p=0.7521
p=0.076
p=0.1973
p=0.1322
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

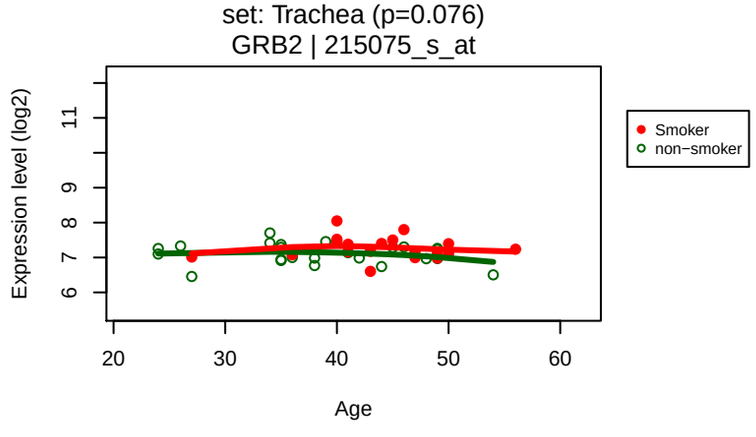

Gene expression ratio in Males
gene: GRB2 | 215075_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6382
p=0.2916
p=0.307
p=0.0964
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRB2 | 215075_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4761
p=0.1267
p=0.1242
p=0.9554
SmallAirway2
LargeAirway2
Trachea
Macrophage2

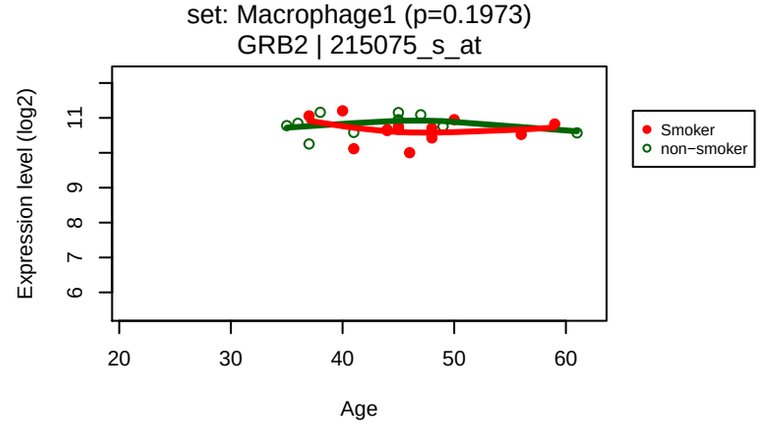

set: SmallAirway1 (p=0.853)
GRB2 | 215075_s_at
Expression level (log2)
Age
Smoker
non−smoker

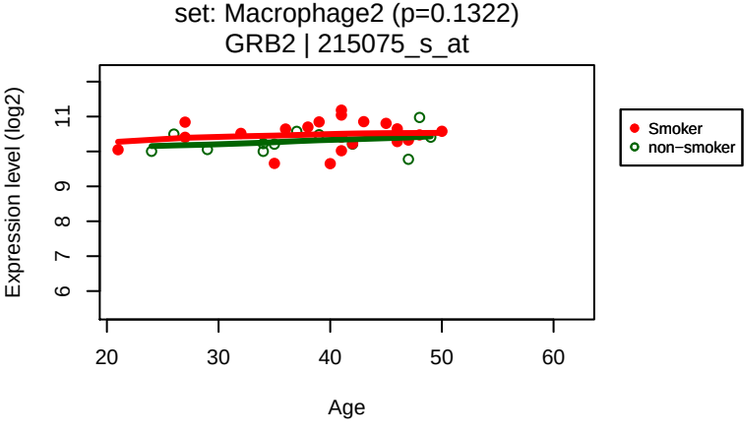

set: SmallAirway2 (p=0.3042)
GRB2 | 215075_s_at
Expression level (log2)
Age
Smoker
non−smoker

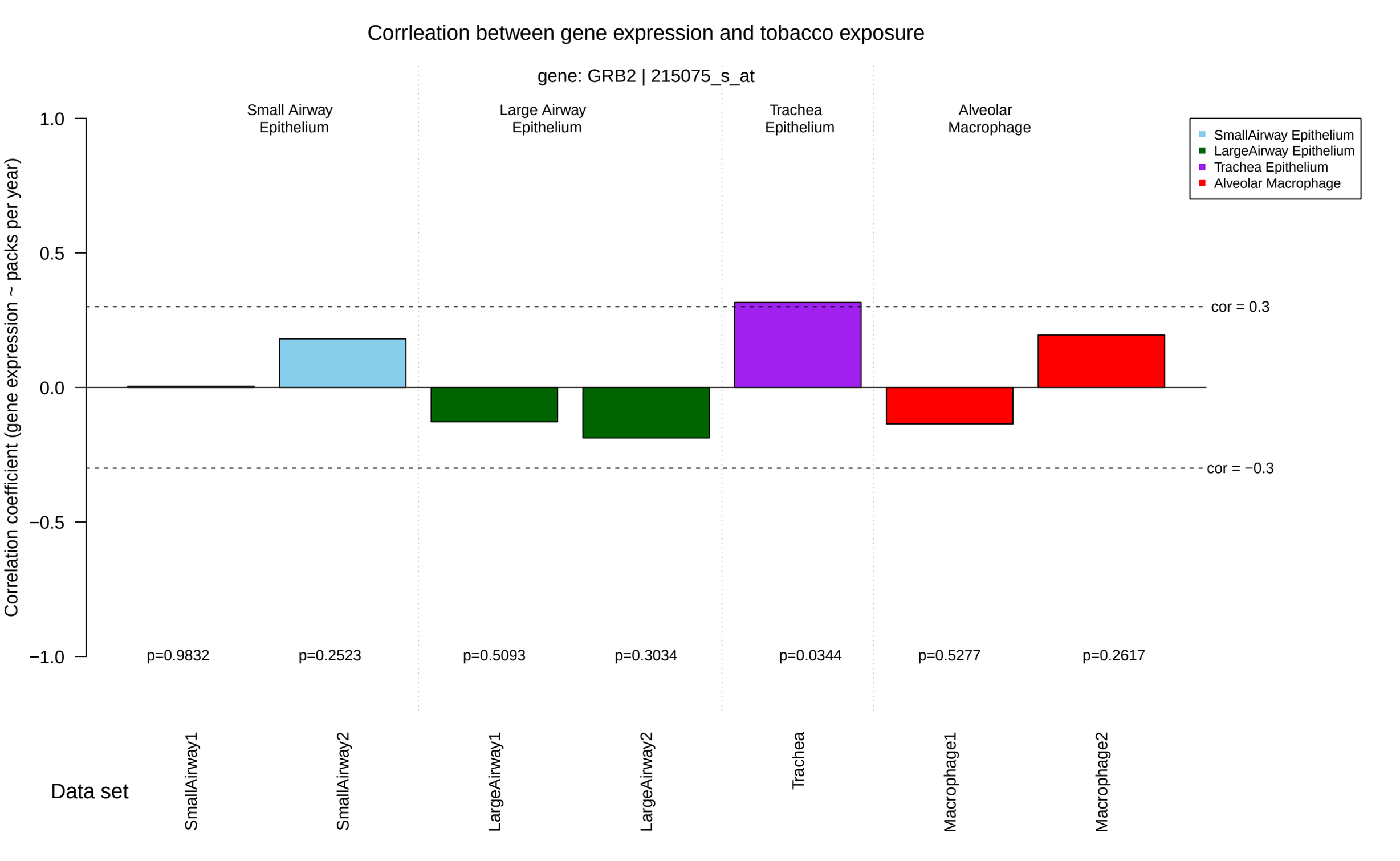

set: LargeAirway1 (p=0.0516)
GRB2 | 215075_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7521)
GRB2 | 215075_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.076)
GRB2 | 215075_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1973)
GRB2 | 215075_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1322)
GRB2 | 215075_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRB2 | 215075_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9832
p=0.2523
p=0.5093
p=0.3034
p=0.0344
p=0.5277
p=0.2617
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2